Comparative Study of Post-Hoc Methods vs Intrinsic Interpretability in Vision Models

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Context and Motivation

The "Black Box" Problem

- Deep learning models (CNNs, ViTs) have opaque decision-making processes.
- **Post-Hoc XAI**: Explainers applied *after* training (e.g., **SHAP** [1], **LIME** [2]).
- **Intrinsic XAI**: Models interpretable by design (e.g., **ProtoPNet** [3]).

Mathematical Foundations: XAI Methods

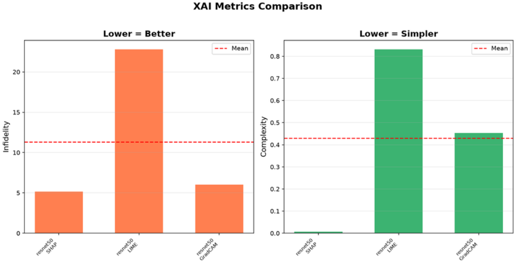
- **SHAP**
  $\phi_i = \sum_S \frac{|S|!(n-|S|-1)!}{n!} \Delta v$
- **LIME**
  $\xi(x) = \arg\min_{g \in G} \mathcal{L} + \Omega(g)$
- **GradCAM**
  $L^c_{\text{Grad-CAM}} = \text{ReLU}\left(\sum_k \alpha^c_k A^k\right)$
- **ProtoPNet**
  $\text{sim} = \log\left(\frac{d^2+1}{d^2+\epsilon}\right)$

Mathematical Foundations: Metrics

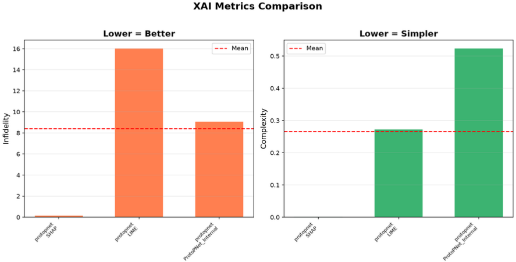
- **Infidelity**
  $\text{INFD} = \mathbb{E}_I \left[\left(f(x) - f(x - I) - I^T \Phi(x) \right)^2 \right]$
- **Complexity (Entropy)**
  $C(\Phi) = -\sum_i p_i \ln(p_i)$
- **Localization (IoU)**
  $\text{Loc} = \frac{|M \cap B|}{|M \cup B|}$
- **Rank Agreement (Spearman)**
  $\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$

Experimental Setup & Metrics

Post-Hoc Metrics



Intrinsic Metrics

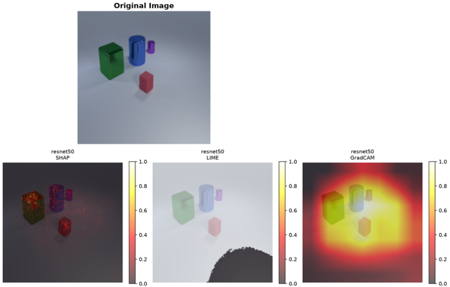


Datasets: *CLE4EVR* (3D objects) and *MNMath* (Handwritten equations).

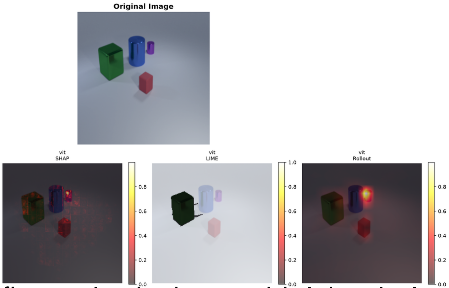
Key Metrics: *Infidelity*, *Complexity*, *Localization*.

Post-Hoc Methods on CLE4EVR Dataset

ResNet50



Vision Transformer (ViT)

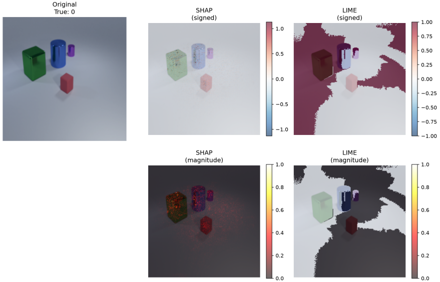


- **SHAP** achieves the lowest complexity, isolating fine-grained edges and bright pixels with high sparsity.
- **GradCAM** performs well: large 3D objects match its coarse receptive field.
- **LIME** underperforms: default SLIC superpixels severely fragment the smooth 3D shapes.

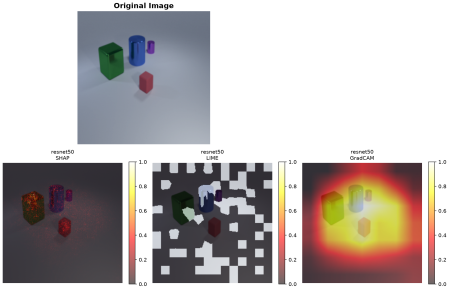
LIME Ablation Study on CLE4EVR

- **Hypothesis**: Does matching LIME's superpixels (K) to the number of objects improve explanations?
- **Result**: Object-aligned boundaries ($K = 4$) unexpectedly double the *Infidelity* compared to default ($K = 50$). Linear surrogates fail to capture the non-linear reasoning of CNNs.

Default LIME ($K = 50$)

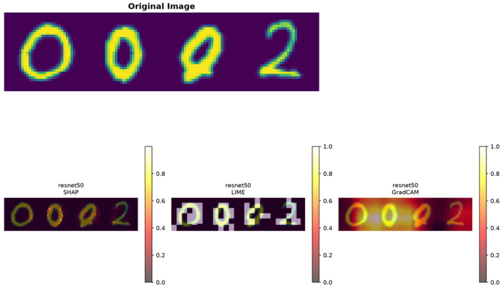


Ablated LIME ($K \approx 4$)

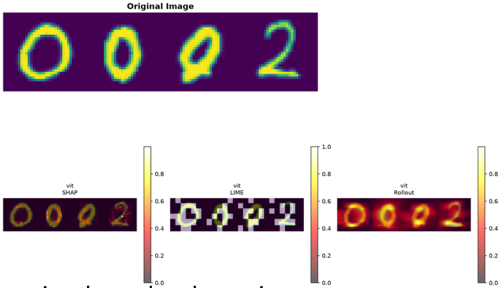


Post-Hoc Methods on MNMath Dataset

ResNet50



Vision Transformer (ViT)

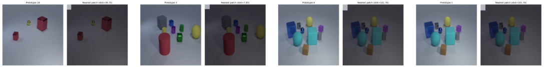


- **SHAP** heavily outperforms GradCAM on fine-grained stroke detection.
- GradCAM's low spatial resolution blurs over the thin handwritten digits.
- **LIME** struggles to isolate thin strokes, as superpixels tend to group the background with the fine handwritten details.

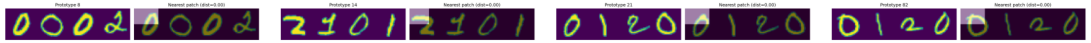
ProtoPNet Learned Prototypes

- ProtoPNet inherently learns a set of visual *prototypes* directly from the training data.
- Classification is based on the distance between the input image patches and these learned prototypes.

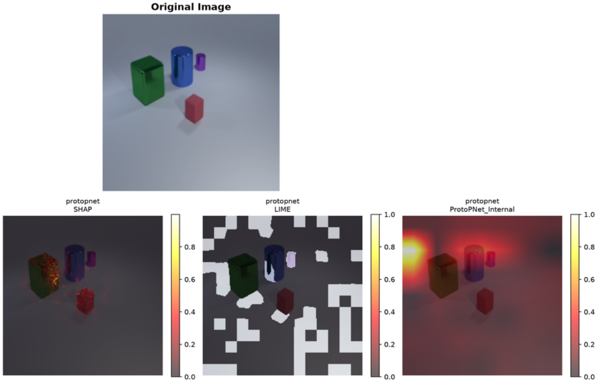
CLE4EVR Prototypes



MNMath Prototypes



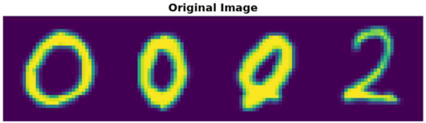
Visual Comparison on CLE4EVR

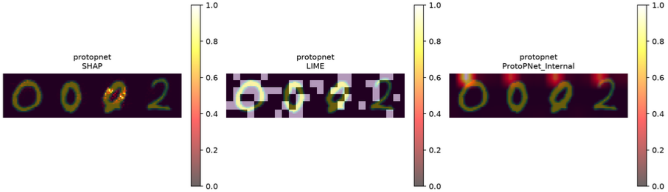


SHAP isolates edges (sparse), while **ProtoPNet Internal** highlights broad, context-aware regions.

Intrinsic vs Post-Hoc: MNMath

Visual Comparison on MNMath



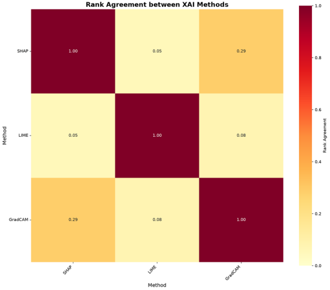


ProtoPNet reasoning encompasses the whole digit and its context, completely ignoring the strict "stroke" pixels favored by SHAP.

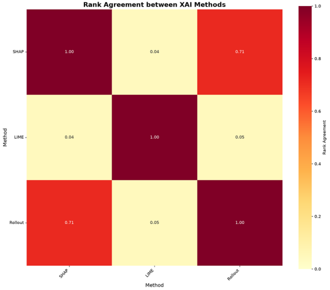
Rank Agreement: CLE4EVR Dataset

- Correlation between Spatial Locality (Intersection) and Infidelity.
- Across all architectures, post-hoc rank correlation is **near zero or negative**.

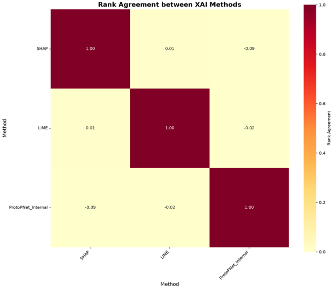
ResNet50



ViT



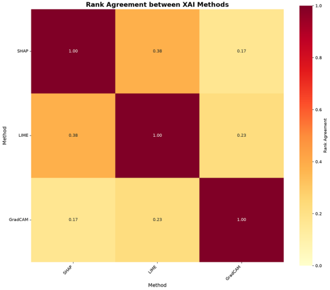
ProtoPNet



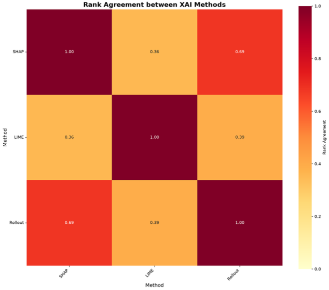
Rank Agreement: MNMath Dataset

- The disconnect is even worse on fine-grained data (-0.24 for ProtoPNet).
- Post-hoc methods misrepresent the actual broad spatial reasoning of the network!

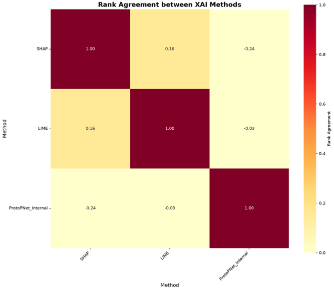
ResNet50



ViT



ProtoPNet



Conclusion

Key Takeaways

- Relying strictly on "Localization Accuracy" is dangerous: it assumes networks reason using human-defined object boundaries.
- Intrinsic interpretability reveals that models learn broad spatial contexts, unlike what post-hoc methods suggest [4].

Thank You!

Questions?

References I

- S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” *Advances in neural information processing systems*, vol. 30, 2017.
- M. T. Ribeiro, S. Singh, and C. Guestrin, “”why should i trust you?” explaining the predictions of any classifier,” in *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*, 2016, pp. 1135–1144.
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- C. Rudin, “Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead,” *Nature Machine Intelligence*, vol. 1, no. 5, pp. 206–215, 2019.